



BRAINWARE UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Lab Assignment of

Course Name: cloud computing Lab

Course Code: pcc-cs591c

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Assignment-1

01

Objective: Install virtualbox/VMware Workstation with different flavours of ~~Windows~~ Linux.

Solution: Different steps to install virtualBox on VM Ware workstation on windows 10/11 and then running different Linux OSes on virtual machines:

Step 1: Download & install virtualization software say Oracle VirtualBox

1. Go to <https://www.virtualbox.org/wiki/Downloads>
2. Download virtualBox for windows hosts.
3. Run the installer and follow the setup wizard.

Step 2: Enable the virtualization in BIOS/UEFI.

1. Reboot your computer and enter BIOS/UEFI. (Usually by pressing Del, F2, or Esc during boot).
2. Look for Intel VT-x/AMD-V and enable it.
3. Save and exit BIOS.

Step 3: Download ISO Files for OS installation popular Linux Distributions;

- Ubuntu: <https://ubuntu.com/download/desktop>
 - Fedora: <https://getfedora.org/>
 - Debian: <https://www.debian.org/distrib>
 - Kali Linux: <https://www.kali.org/get-kali/>
 - Linux Mint: <https://linuxmint.com/download.php>
- windows ISOs:
- Windows 10: <https://www.microsoft.com/software-download/windows10>
 - Windows 11: <https://www.microsoft.com/software-download/windows11>

Step 4: create virtual Machine

1. Open VirtualBox and click ~~New~~ New.
2. Enter VM name, select type (Linux/windows) and version.
3. Allocate RAM (e.g. 2048MB or more)
4. Create a virtual hard disk (VHD, dynamically allocated - 20GB+).

5. Select the VM → Settings → Storage → click on "Empty" under controller: IDE.
6. click the disk icon → choose a disk file → select your ISO.
7. Start the VM and proceed with OS Installation
8. Install Guest Additions/VMware Tools.
- For Better Performance (mouse integration, screen resizing, etc)
- VirtualBox : Devices → Inserted Guest Additions CD Image → select VBoxGuestAdditions.iso file from C:\Program Files\Oracle\VirtualBox

Example configurations OS Flavour	Suggested RAM	Disk Space	Notes
Ubuntu Desktop	2-4 GB	25 GB	Great for development.
Kali Linux	2-4 GB	20 GB	Security testing.
Windows 10	4-8 GB	30+ GB	Full Windows experience.
Linux Mint	2 GB	20 GB	Lightweight and beginner-friendly.

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Assignment - 2 & 3

03

Objective: To install a C-compiler in the virtual machine created using virtual box and execute simple programs.

Solutions: The different steps are -

Step-1: Start your Linux virtual Machine Launch your VM (Ubuntu, Debian, etc) in Virtual Box.

Step-2: Open the Terminal.

you can press

ctrl + Alt + T

Or go to Applications → Terminal.

Step-3: Update the Package List

sudo apt update.

Step-4: Install gcc (GNU compiler collection).

sudo apt install build-essential.

This installs:

- gcc: C compiler
- g++: C++ compiler
- make: Build utility
- Header files: (libc-dev, etc)

Verify installation:
gcc -version

Step-5: Write a simple C program

Option-1: Using nano or vim in terminal

nano hello.c

Paste this code:

```
#include <stdio.h>
int main() {
    printf("Hello, VirtualBox!");
    return 0;
}
```

Save and exit:

For nano: press ctrl + O, then Enter, then ctrl + X

Step 6:

compile the c program
gcc hello.c -o hello

This creates an executable named hello.

Step 7:

Run the Executable.

./hello

Assignment-3

Extra: Another program

```
#include <stdio.h>
```

```
int main () {
```

```
    int a, b, sum;
```

```
    printf ("Enter two integers: ");
```

```
    scanf ("%d %d", &a, &b);
```

```
    sum = a + b;
```

```
    printf ("sum = %d", sum);
```

```
    return 0;
```

}
Save as sum.c - compile as gcc sum.c -o sum

./sum

output:

Enter two integers: 15 20

sum: 35

Assignment - 4

05

1) Install Google App Engine. Create hello world app

Step 1: Install Prerequisites

Open our VM's terminal and run:

`sudo apt update`

`sudo apt install python3 python3-pip git -y`

Step 2: Install Google Cloud SDK

Add Google Cloud APT Repo and GPG (GNU Privacy Guard) key

Step 3: Update and install the SDK

`sudo apt update`

`sudo apt install google-cloud-sdk`

Step 4: Initialize Google Cloud SDK

Run:

`gcloud init`

- Sign in to our Google account
- Choose or create a GCP project
- Set a region if prompted.

Step 5: Install App Engine Component

`sudo apt-get install google-cloud-sdk-app-engine-python`

Step 6: Create Hello World App (Python)

1. create a project directory

2. create main.py

3. create app.yaml

Step 7: Install Dependencies

1. Create Virtual Environment

2. `python3 -m venv`

3. Activate Virtual Environment

4. Install Packages from requirements.txt

Step 8: Deploy to Google App Engine

Step 9: Before deployment:

a) Check: `gsutil ls`

b) Verify the Environment, make sure you are deploying to the right project, cloud config list

Step 3: Open the App in Browser
getcloud app browse

Oh, Run your app locally

pip install -b requirements.txt

gunicorn -b 0.0.0.0:8080 main.py

OK

Assigning Storage Permission

Click the pencil (edit) icon next to that service account now.

now, 1. Click "Add Another Role"

2. In the Role Selector, search and select:

Storage Admin

(on minimum: Storage Object Admin)

3. Click Save

N.B. It is seen that for using pip - command, we must set env - environment variable as -

a) Within the project - folder
python3 -m venv venv

b) Activate Virtual Environment
source venv/bin/activate

Some Terms:

- 1) -y : yes
- 2) -b : bind and it tells gunicorn where to listen.
- 3) GPG : GPG stands for GNU Privacy Guard
- 4) Git : helps track changes in files
- 5) m : Module to run a Python module as a script


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Assignment-45

Implementing python setup & simple programming in VM-Ware.

⇒ Sudo apt updated
Sudo apt install vim
y

Vim hellow.py

n=int(input("Enter the number of terms:"))

a,b = 0,1

print("Fibonacci Series:")

for i in range(n):

print(a, end = " ")

≡;wq a,b = b, a+b

Output:

python3 hellow.py

Enter the number of terms: 10

Fibonacci Series:

0 1 1 2 3 5 8 13 21 34

2) Write a python code for counting words/tokens in a string as given.

⇒ text = input("Enter a string:")

words = text.split()

count = len(words)

print("Tokens/Words:", words)

print("Number of Words/Tokens:", count)

≡;wq

Output:

Vim count.py

python3 count.py

Enter a string: Hello I am Shibnath Sahoo

Tokens/Words: ['Hello', 'I', 'am', 'Shibnath', 'Sahoo']

Number of Words/Tokens: 5

Assignment-6

Q8)

- 1) Write a python program in VM Ware to add two numbers using user input.

```

⇒
n1 = int(input("Enter first number:"))
n2 = int(input("Enter second number:"))
n3 = n1 + n2
print(f"Addition of two numbers {n1} and {n2} is: {n3}")

```

Output:

```

vim addition.py
python3 addition.py
Enter first number: 8
Enter second number: 10
Addition of two numbers 8 and 10 is: 18

```

- 2) Write a C program in VM Ware to add two numbers using user input.

```

⇒ #include <stdio.h>
int main() {
    int a, b, s;
    printf("Enter first number:");
    scanf("%d", &a);
    printf("Enter first second number:");
    scanf("%d", &b);
    s = a + b;
    printf("Addition of %d and %d is %d", a, b, s);
    return 0;
}

```

Output:

```

Enter first number: 5
Enter second number: 3
Addition of 5 and 3 is 8

```


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1. GitHub Account Creation :-

- Step 1: Go to <http://github.com/join> in used browser.
- Step 2: Enter the required personal details like user name, Email address and password.
- Step 3: click verify to start the verification puzzle.
- Step 4: After the verification click on the green create account button.
- Step 5: Then you will receive a verification code in your gmail, put that code in verification page.
- Step 6: Select your preference if you want to and then click continue or just press skip personalization.
- Step 7: No. to the type of plans by GitHub (Free, pro, Team, Enterprise).
- Step 8: After selecting the plan your account will be created.

Creating a HelloWorld Python code in github.

- Step 1: Create a new repository and select owner followed by repository name.
- Step 2: Select the visibility (public or private) then click on create repository.
- Step 3: Now click on Add file and select create new file.
- Step 4: Enter file name followed by .py and write the Python code in the editor.
- Step 5:

```
print("Hello World")
```

 Now click on commit changes.

- Step 6: ~~Open~~ Open the repository and click on code and copy the HTTPS clone Link.

Set up Google cloud console :-

- Step 1: Go to console.cloud.google.com and login with your google account.
- Step 2: Create a new project and give it a name.

#Run Python code from Git Repository:-

Step 1: In the Google cloud console, navigate to cloud shell from the top right.

Step 2: clone your GitHub repository using git clone <https://github.com/05A05746S/Hello.git>

Step 3: Navigate to your project folder using

Step 4: `cd Hello`

Type `ls` to view file name.

Step 5: Run python code using `python hello.py`.

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Assignment-8

and multiply.

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1) Creating a Python program to add ^{and multiply} two matrices in Github

Step 1: Create a new repository and select owner followed by repository name.

Step 2: Select visibility (Public or private) then click on the create repository.

Step 3: Now click on Add file and select create new file

Step 4: Enter file name followed by .py and write python code in the condition.

list1 = [(1,2,3), (4,5,6)]

list2 = [(7,8,9), (10,11,12)]

add = [a+b for ab in zip(sublist1, sublist2) for
sublist1, sublist2 in zip(list1, list2)]

print(add)

Step 5: Setup Google cloud console (GCE)

Step 1: In GCE Navigate to cloud shell from the top right.

Step 2: clone your github repository using
git clone <https://github.com/SA57465/Multiply.git>

Step 3: Navigate to your project folder using cd Multiply

Step 4: Run Python code using Multiply.py

O/P: [(8,10,12), (14,16,18)]

Multiply:

[27, 30, 33]

[78, 87, 96]

product = [sum(a*b for a,b in zip(row,col)) for row in list1]

for row in zip(*~~list1~~list2)] for row in list1]

print("Multiply")

for row in product:

print(row)

Assignment - 2

- 1) Write a program in python to find factorial of a number using github and google cloud console.

```
⇒ def factorial(n):  
    if n == 0 or n == 1:  
        return 1  
    else:  
        return n * factorial(n-1)  
  
print(factorial(5))
```

Output:

```
git clone https://github.com/SA57465/factorial.git  
ls  
Factorial Hello Multiply  
cd factorial  
ls  
Factorial factorial.py  
python fact factorial.py  
120
```

- 2) Write a program in python to check a string is palindrome or not using github and google console.

```
⇒ def is_palindrome(s):  
    s = ''.join(c.lower() for c in s if c.isalnum())  
    return s == s[::-1]  
  
word = input("Enter a string;")  
if is_palindrome(word):  
    print("It's a palindrome!")  
else:  
    print("Not a palindrome!")
```

Output:

```
git clone https://github.com/SA57465/factorial.git.  
ls  
Factorial factorial.py  
cd factorial  
ls  
factorial.py palindrome.py  
Enter a string; mam  
It's a palindrome!
```


Assignment-10

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1) Write a Java program to check a number is palindrome or not ~~using~~ in github.

```

=> import java.util.Scanner;
public class PalindromeNumber {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number:");
        int number = scanner.nextInt();
        if (isPalindrome(number)) {
            System.out.println(number + " is a
                                palindrome.");
        }
        else {
            System.out.println(number + " is not a
                                palindrome.");
        }
        scanner.close();
    }
    public static boolean isPalindrome(int num) {
        int originalNumber = num;
        int reversedNumber = 0;
        while (num != 0) {
            int digit = num % 10;
            reversedNumber = reversedNumber * 10 +
                                digit;
            num /= 10;
        }
        return originalNumber == reversedNumber;
    }
}

```

Output:

```

git clone https://github.com/SA574645/Hello.git
ls
Hello hello.py
cd Hello
ls
hello.py multiply palindrome.java
Java palindrome.java
Enter a number: 121
121 is a palindrome.

```

2) Write a Java Program to make a simple calculator calculator using switch case in github.

```

import java.util.Scanner;
public class SimpleCalculator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter first number:");
        double num1 = scanner.nextDouble();
        System.out.print("Enter second number:");
        double num2 = scanner.nextDouble();
        System.out.println("Choose an operation:");
        System.out.println("1 - Addition (+)");
        System.out.println("2 - Subtraction (-)");
        System.out.println("3 - Multiplication (*)");
        System.out.println("4 - Division (/)");
        System.out.print("Enter your choice:");
        int choice = scanner.nextInt();
        double result;
        switch (choice) {
            case 1:
                result = num1 + num2;
                System.out.println("Result: " + result);
                break;
            case 2:
                result = num1 - num2;
                System.out.println("Result: " + result);
                break;
            case 3:
                result = num1 * num2;
                System.out.println("Result: " + result);
                break;
            case 4:
                if (num2 != 0) {
                    result = num1 / num2;
                    System.out.println("Result: " + result);
                } else {
                    System.out.println("Error: Division by zero is undefined.");
                }
                break;
            default:
                System.out.println("Invalid choice. Please select 1, 2, 3, or 4.");
                break;
        }
        scanner.close();
    }
}

```

Output: git clone ~~https://github.com/~~ ~~simple-calculator~~ pull

```

Java calculator.java
Enter first number: 5
Choose an operation:
1 - Addition (+)
2 - Subtraction (-)
3 - Multiplication (*)
4 - Division (/)
Enter your choice: 1 Result: 5.0

```

Handwritten signature
15/09/25